**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 047 · Inputs, Processes and Outputs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is algorithmic thinking?

Q2. What does an encryption utility produce?
A. A smaller file
B. A sorted folder
C. Ciphertext
D. Extra storage space

Q3. What does a compression utility do?

Q4. A CPU reads an instruction at memory address 120. Correct both claims. Explain why. (a) MAR holds a calculation result. (b) MDR always holds the next instruction's address.

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# I Do · Read new knowledge

## Key vocabulary

Input — Data supplied to a program.

Processing — Operations or decisions performed on data.

Output — Result produced by an algorithm.

IPO — Input–Process–Output model.

Algorithm — Instructions for a problem-solving task.

Sensor — Device producing measured input.

Threshold — A comparison value triggering an outcome.

## New learning

Start with the result: Every problem asks for a useful result: a total, decision, message or control signal. Identify what the result should be first.

Identify specific inputs: Inputs are supplied values such as length and width, a sensor reading or a selected option. Name exact values, rather than saying 'some data'.

Describe the processing: Processing includes calculations, comparisons or decisions. For example: total = unitPrice × quantity, rather than 'the computer works it out'.

Output completes the task: Output could be a displayed cost or warning message. Check that it follows from the processing and answers the stated question.

## Worked example

A program takes a rectangle's length and width and calculates its area. Identify the input, processing and output. [4 marks]

1. The result required is the rectangle's area.

2. Inputs are length and width [1], two values supplied to the algorithm [1].

3. Processing multiplies length × width [1].

4. Output displays the area [1]. A separate result follows the calculation.

# We Do · Practise with guidance

### Guided 1

A snack shop program calculates the total for five identical snacks. Identify input, processing and output. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: Inputs: price per snack [1] and quantity 5 [1]. Process: multiply price × 5 [1]. Output: the total price [1].

**Guided 2**

A temperature monitor displays an ice warning when a reading is below 2°C. Explain input, processing and output. [4 marks]

Worked method: Temperature from the sensor is input [1]. The program compares it with 2°C [1]. If below the threshold [1], it outputs an ice warning [1].

**Guided 3**

A ticket machine accepts an adult fare and a child fare and displays their total. Give four ordered steps. [4 marks]

Worked method: Input adult fare [1]. Input child fare [1]. Add them [1]. Display the sum [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Write the line that prints the pixel count.

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
```

write

## Question 1

Define processing within an algorithm. [2 marks] Source: Original J277-style teaching question, not an official past paper.

## Question 2

A streaming service calculates subscription cost from months purchased and a price per month. Describe the IPO stages. [4 marks] Source: Original J277-style teaching question, not an official past paper.

## Question 3

Explain why identifying IPO accurately helps when designing a school uniform ordering algorithm. [6 marks] Source: Original J277-style teaching question, not an official past paper.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Exit Ticket · Answer before checking

Output is produced after suitable processing has been performed on required inputs.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Developing a clear, ordered and unambiguous method for solving a problem.

Q2. C — Ciphertext

Q3. It reduces the storage space required for data, using a suitable lossless or lossy method.

Q4. (a) The MAR holds the address being accessed. (b) The MDR holds data or an instruction moving to or from memory. An address is different from the data at it.

## I Do / We Do pages · Guided practice

Q1. Inputs: price per snack [1] and quantity 5 [1]. Process: multiply price × 5 [1]. Output: the total price [1].

Q2. Temperature from the sensor is input [1]. The program compares it with 2°C [1]. If below the threshold [1], it outputs an ice warning [1].

Q3. Input adult fare [1]. Input child fare [1]. Add them [1]. Display the sum [1].

## You Do pages · Code First

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
print(pixels)
```

Expected result: 80

## You Do pages · Independent questions

Q1. Operations/calculations performed on inputs [1] to obtain results [1].

Q2. Inputs are number of months [1] and monthly price [1]. Processing multiplies them [1]. Output displays the cost [1].

Q3. Inputs could be item types and quantities [1] and unit prices [1]. Processing multiplies unit prices and quantities [1] and totals the order [1]. Output gives total price/confirmation [1]. Precise IPO helps plan/test the algorithm [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**